

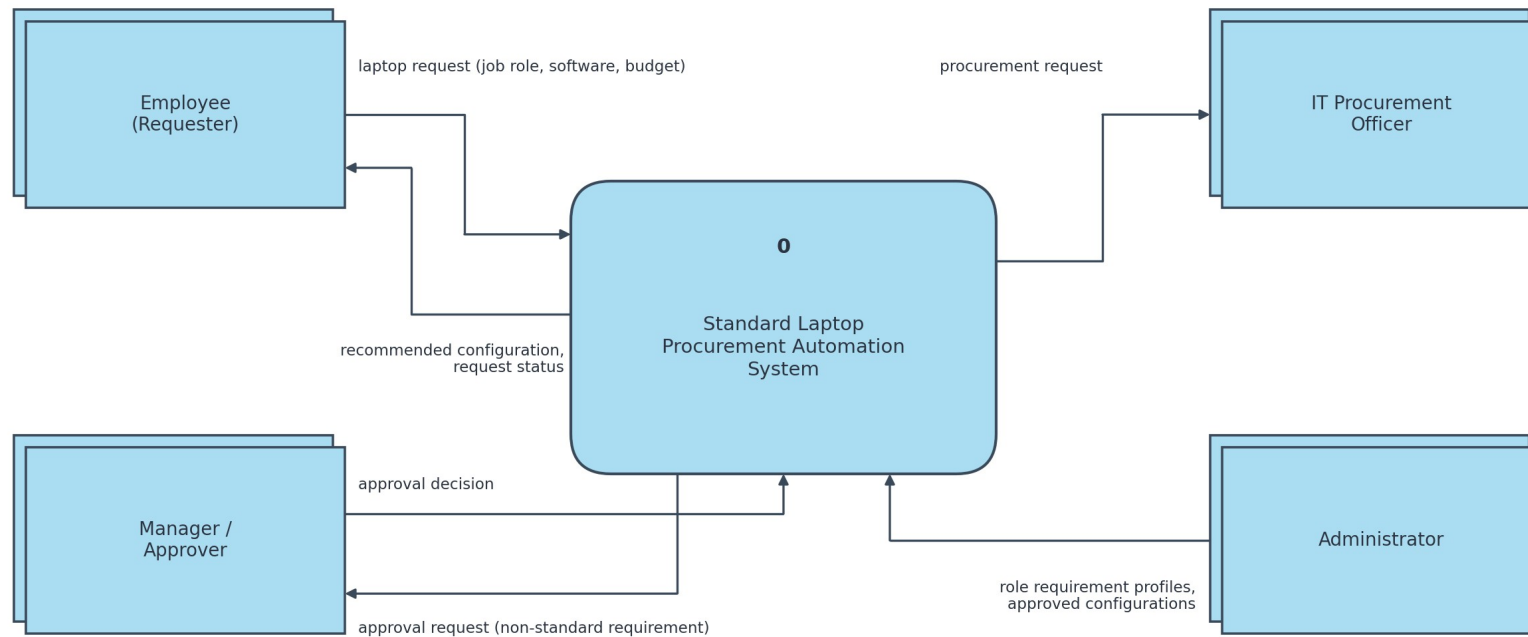
Project Design Phase-II
Data Flow Diagram & User Stories

Date	02 October 2026
Team ID	SWTID-2026-4912
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Diagram)

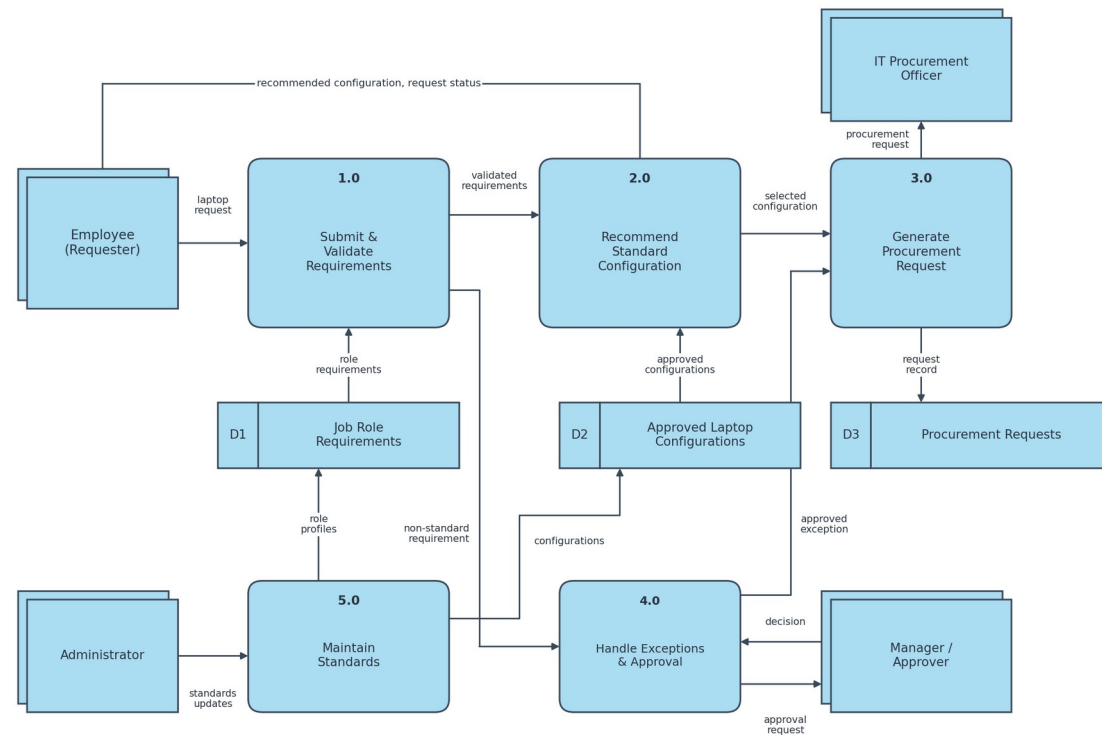


Level 0 flows: the Employee (Requester) submits a laptop request with job role, required software and budget; the system returns a recommended standard configuration and the request status.

Non-standard requirements are sent to the Manager / Approver as an approval request; the approval decision is returned to the system.

The IT Procurement Officer receives the generated procurement request, and the Administrator supplies role requirement profiles and approved laptop configurations.

DFD Level 1



1. The Employee submits a laptop request (job role, software, budget). Process 1.0 validates it against the Job Role Requirements (D1).
2. Validated requirements go to Process 2.0, which matches them with the Approved Laptop Configurations (D2) and returns a recommended configuration to the Employee.
3. Requirements that differ from the standard are flagged as non-standard and passed to Process 4.0 for exception handling and approval by the Manager / Approver.
4. Process 3.0 generates the procurement request from the selected configuration or an approved exception, stores it in Procurement Requests (D3) and sends it to the IT Procurement Officer.
5. The Administrator maintains role profiles and approved configurations through Process 5.0, which updates D1 and D2.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Employee (Requester)	Requirement Submission	USN-1	As an employee, I can submit a laptop request by entering my job role, required software and budget limit.	Request is saved and a request ID is displayed	High	Sprint-1
		USN-2	As an employee, I can select my job role from a predefined list so that the standard requirements for that role are applied automatically.	Standard CPU, RAM, storage and GPU requirements for the role are loaded	High	Sprint-1
	Requirement Validation	USN-3	As an employee, I am notified if my request has missing or invalid details.	Invalid fields are highlighted and the request is not submitted until corrected	High	Sprint-1
		USN-4	As an employee, I am informed when my request contains a non-standard requirement.	Request is flagged as non-standard and the differing requirement is shown	High	Sprint-2
	Laptop Recommendation	USN-5	As an employee, I receive a recommended standard laptop configuration based on my job role, required software and budget.	Recommendation lists CPU, RAM, storage and GPU, and is within the budget	High	Sprint-2
		USN-6	As an employee, I can review and confirm the recommended configuration.	Confirmation is recorded and the request moves to procurement request generation	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
	Request Tracking	USN-7	As an employee, I can track the status of my laptop request.	Current status is visible (Submitted, Pending Approval, Approved, Rejected, Sent to Procurement)	Medium	Sprint-3
IT Procurement Officer	Procurement Request Generation	USN-8	As an IT procurement officer, I receive an automatically generated procurement request for each confirmed configuration.	Request contains requester, job role, configuration, required software and budget	High	Sprint-2
		USN-9	As an IT procurement officer, I can view and filter the list of procurement requests by status.	List can be filtered by status and shows the latest requests first	Medium	Sprint-3
Manager / Approver	Exception Handling & Approval	USN-10	As a manager, I receive an approval request when an employee's requirement is non-standard.	Request shows the requested and standard specifications for comparison	High	Sprint-3
		USN-11	As a manager, I can approve or reject an exception request and add a comment.	Decision is recorded and the employee and IT procurement officer are notified	High	Sprint-3
Administrator	Standards Management	USN-12	As an administrator, I can create and update job role requirement profiles (CPU, RAM, storage, GPU, software and budget).	Updated profiles are used for all new requests	High	Sprint-1
		USN-13	As an administrator, I can add, edit or retire approved laptop configurations.	Changes apply to all subsequent recommendations	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
	Exception Handling & Approval	USN-14	As an administrator, I can view the history of exception requests and their decisions.	History lists each exception with requester, decision and date	Medium	Sprint-3